

1.

The winter sale began, and Mrs. Sarah wants to buy a T-shirt for her son. The T-shirt has a ticket that indicates a discount of $X\%$ and its price P after the discount. She now wants to know the price before the discount to know if the discount was worthy. Help her!

Input

Only one line containing two numbers X and P ($1 \leq X \leq 99, 1 \leq P \leq 4 \times 10^4$) – the discount percentage, and the price of the T-shirt after the discount.

Output

Print the price of the T-shirt before the discount rounded up to **two decimal places**.

Examples

input	Copy
20 80	
output	Copy
100.00	

input	Copy
15 20	
output	Copy
23.53	

2.

Given a lowercase alphabet character. You have to print the next character in the alphabet.

Input

Only one line containing a lowercase English letter C .

Output

Print the next letter to C in the alphabet.

Example

input	Copy
a	
output	Copy
b	

Note

The next letter to **z** is **a**.

3.

Given two numbers a and b . You have to answer with "YES" if there is a **non-empty interval** consisting of numbers from l to r ($l, l+1, l+2, \dots, r$) with **a** odd numbers and **b** even numbers, or "NO" otherwise.

Input

Only one line containing two numbers a and b ($0 \leq a, b \leq 100$) the number of odd numbers and the number of even numbers in the interval respectively.

Output

Print "YES" or "NO" as described in the statement.

Examples

input	Copy
2 3	
output	Copy
YES	

input	Copy
3 1	
output	Copy
NO	

4.

The Egyptian football team will be in Russia for the World Cup. Of course, they all would like to buy souvenirs for their families. Luckily, they met the king of souvenirs Matryoshka who is famous for his masterpiece Katryoshka. He makes it using different wooden pieces: **eyes**, **mouths** and **bodies**. He can form a nice Katryoshka using one of the following combinations:

1. Two eyes and one body.
2. Two eyes, one mouth, and one body.
3. One eye, one mouth, and one body.

If the king has n eyes, m mouths and k bodies, what is the largest number of Katryoshkas he can make?

Input

Only one line containing three numbers n, m and k ($0 \leq n, m, k \leq 10^{18}$) – the number of eyes, mouths and bodies respectively.

Output

Print the largest number of Katryoshkas he can make.

Examples

input	Copy
1 2 3	
output	Copy
1	

input	Copy
0 11 2	
output	Copy
0	

input	Copy
90 24 89	
output	Copy
57	

5.

A number of two digits is lucky if **one of its digits is divisible by the other**.

For example, 39, 82, and 55 are lucky, while 79 and 43 are not.

Given a number between 10 and 99, determine whether it is lucky or not.

Input

Only one line containing a single number N ($10 \leq N \leq 99$).

Output

Print "YES" if the given number is lucky, otherwise print "NO".

Examples

input	Copy
39	
output	Copy
YES	

input	Copy
64	
output	Copy
NO	